

PHOENIX – Trimester 2 Cybersecurity

Project: PHOENIX

Company: Hardhat Enterprises

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1. Purpose

The purpose of this document is to review the cybersecurity work inherited from Trimester 1 and identify the main cybersecurity priorities that should be continued during Trimester 2 of Project PHOENIX.

As a returning member of PHOENIX and the Cybersecurity Stream Leader for Trimester 2, it is important for me to understand what has already been completed before planning new work. This helps avoid repeating previous work and allows the cybersecurity stream to focus on areas that still require implementation, integration, testing, or improvement.

This review also provides a starting point for cybersecurity sprint planning and my individual technical contribution during the trimester.

2. PHOENIX Cybersecurity Context

PHOENIX is a disaster-cyber resilience platform designed to combine disaster information, cybersecurity threat intelligence, AI/ML risk modelling, backend services, and user-facing interfaces.

The cybersecurity stream has an important role because emergency and disaster situations can create opportunities for cyber threats such as:

- Phishing and malicious links
- Fake emergency alerts
- Misinformation

- Unverified information sources
- Data tampering
- API abuse
- Unauthorised access
- Privilege abuse
- Denial-of-service attacks
- Data breaches and ransomware

Cybersecurity therefore needs to be considered across the PHOENIX system rather than being treated as a separate component.

3. Trimester 1 Cybersecurity Work

The Trimester 1 handover shows that the cybersecurity stream progressed from mainly planning and documentation towards implementation, integration, and security testing.

Major areas completed or progressed included:

3.1 Secure System Design

Security requirements and controls were developed for different parts of PHOENIX.

Existing security design work includes areas such as:

- Authentication
- Authorisation
- Role-based access control
- Input validation
- Rate limiting
- Secure API design
- Digital signature verification
- Logging
- Data validation

The PHOENIX repository also contains secure-design documentation relating to input validation, rate limiting, signing workflows, and verification processes.

These existing controls should be reviewed and reused where appropriate rather than redesigned without considering previous work.

4. Threat Analysis and Risk Assessment

Threat analysis and risk modelling were significant areas of the Trimester 1 cybersecurity work.

Cybersecurity threats relevant to PHOENIX were researched and analysed based on their likelihood, impact, and possible consequences.

The existing risk model supports:

- Likelihood scoring
- Impact scoring
- Risk categorisation
- Threat prioritisation
- Risk assessment documentation

My own Project A contribution focused heavily on this area.

I worked on CY019 Risk Assessment / Risk Scoring and helped develop a structured approach for converting cybersecurity and AI-related outputs into a risk assessment result.

My work included concepts such as:

- Legitimacy classification
- Likelihood
- Impact
- Final risk score
- Risk percentage
- Risk level
- Matched cybersecurity threats
- STRIDE threat categories
- Recommended security controls

I also developed an API response schema and JSON structure to show how these results could eventually be communicated between the AI/ML, backend, frontend and cybersecurity components of PHOENIX.

This previous work can support Trimester 2 integration activities because risk information needs to be structured consistently when it moves between different PHOENIX components.

5. Existing Security Testing

Security testing was also completed against parts of PHOENIX during Trimester 1.

Testing areas included:

- Authentication attack testing
- Unauthenticated access testing
- Data-ingestion endpoint testing
- Hazard endpoint testing
- Threat-data access testing
- API response verification
- OWASP ZAP baseline scanning

Tools used across the cybersecurity testing work included:

- Postman
- Burp Suite
- OWASP ZAP
- SQLMap
- Python-based testing scripts

Several security concerns were identified during this work, including:

- Missing rate limiting on some authentication workflows
- Missing logout functionality in some backend areas
- Client-controlled role fields
- Unauthenticated access to some ingestion endpoints
- Potential unauthenticated threat-data access

These findings are important for Trimester 2 because security testing should not stop when a vulnerability is identified. Where relevant to the current system, findings should be checked against the latest implementation, remediated, and tested again.

6. Threat Detection and Response

Threat detection and response functionality also progressed during Trimester 1.

Existing API functionality included threat-related endpoints and threat-ingestion capabilities.

Testing was performed using examples including:

- Ransomware events
- Phishing events
- Malware-related data

Validation included threat ingestion, threat response mapping, threat statistics generation and API response verification.

This provides an existing cybersecurity foundation that can potentially be connected with other PHOENIX components as the system becomes more integrated.

7. TEAVS and ADCRS Integration

One of the areas carried forward from Trimester 1 is the integration between TEAVS and ADCRS.

According to the project handover, previous work in this area included:

- Research into the interaction between TEAVS and ADCRS
- Analysis of secure data flow between the systems
- Example logic-flow diagrams
- A Python prototype using dummy data
- Documentation of secure communication requirements
- Identification of potential threats and mitigation strategies

However, full implementation was not completed because access and some required system components were not ready.

The recommended continuation was to:

- Continue integration when the required backend components are available
- Implement live secure communication between TEAVS and ADCRS
- Apply secure token-based communication and encryption
- Conduct integration testing
- Conduct threat and security validation

This is particularly relevant to my Trimester 2 work because **Implement TEAVS – ADCRS Integration** has been assigned as one of my individual technical responsibilities.

Before implementation begins, the existing TEAVS and ADCRS documentation, code, prototype and dependencies need to be located and reviewed.

The exact implementation approach should only be decided after the current architecture and available components are understood.

8. Governance and Compliance

Trimester 1 also produced cybersecurity governance and compliance work.

This included work aligned with frameworks and standards such as:

- NIST Cybersecurity Framework
- ASD Essential Eight
- ISO 27001

Existing work covered areas such as:

- Data protection
- Security governance
- Risk management
- Incident response
- Privacy
- Responsible data handling
- Security guidelines
- Cyber resilience

These documents provide useful guidance for Trimester 2, but governance requirements should continue to be checked as the technical architecture changes.

9. Current Cybersecurity Gaps and Remaining Work

Based on the Trimester 1 handover, several important cybersecurity areas remain incomplete or require further development.

Priority 1 – TEAVS and ADCRS Integration

The existing research and prototype work needs to be reviewed and, where technically possible, progressed towards actual integration.

Priority 2 – Security Testing and Remediation

Existing security findings should be reviewed against the current system. New or modified functionality should also undergo appropriate security testing.

Priority 3 – Monitoring and Logging

More advanced security monitoring and logging remain areas for further development.

Priority 4 – Automated Security and Integration Testing

Where possible, repeatable tests should be introduced so that important security controls can be checked when the system changes.

Priority 5 – Production and Deployment Security

As PHOENIX moves closer to deployment, authentication, authorisation, secrets, configuration, exposed services and other deployment-related security controls will require validation.

Priority 6 – Cross-Stream Integration

Cybersecurity needs to remain connected with:

- Backend
- Frontend
- AI/ML
- Alert/mobile functionality
- Database and infrastructure components

Security requirements should therefore be communicated during development rather than reviewed only after integration is complete.

10. Trimester 2 Cybersecurity Leadership Priorities

As Cybersecurity Stream Leader, my responsibilities extend beyond my individual TEAVS–ADCRS contribution.

My main leadership priorities for the trimester will be:

Cybersecurity Sprint Planning

Help organise cybersecurity activities into clear and achievable work items and identify important dependencies or blockers.

Threat-Modelling Coordination

Help ensure new integrations and important system changes are considered from a threat perspective and that identified risks are connected with realistic mitigations.

Security Research

Support research where the team needs to understand an unfamiliar security problem, technology, vulnerability, or control before implementation.

Testing Oversight

Help coordinate security testing activities and ensure findings are clearly documented, communicated, and followed up where appropriate.

Contribution Review

Review cybersecurity contributions where required and provide feedback to help keep technical work aligned with the wider security objectives of PHOENIX.

Cross-Team Coordination

Communicate with backend, frontend, AI/ML and other relevant streams when cybersecurity requirements or dependencies affect their work.

11. My Individual Technical Focus

Alongside my leadership responsibilities, my main planned individual technical contribution is:

TEAVS – ADCRS Integration

The immediate objective is not to assume how integration should work.

The first stage will be to locate and understand:

- Existing TEAVS implementation
- Existing ADCRS implementation
- Previous integration research
- Existing Python prototype
- Current API or interface requirements
- Data exchanged between the components
- Authentication requirements
- Current backend dependencies
- Existing security controls
- Technical blockers

After this review, a more detailed integration and security design can be created based on the actual PHOENIX architecture.

This will provide a stronger foundation for implementation, threat modelling and security testing later in the trimester.

12. Immediate Next Actions

The immediate cybersecurity actions following this handover review are:

1. Review the current PHOENIX repository structure and relevant cybersecurity artefacts.
2. Locate existing TEAVS and ADCRS documentation, source code and previous prototype work.
3. Review outstanding cybersecurity work from the Trimester 1 handover.
4. Identify current cybersecurity of team members and their assigned areas.
5. Establish cybersecurity priorities and dependencies for upcoming sprint work.
6. Identify which security findings from Trimester 1 remain relevant to the current system.
7. Begin documenting the current TEAVS–ADCRS architecture and integration requirements.
8. Coordinate with relevant technical streams where TEAVS–ADCRS dependencies exist.
9. Maintain GitHub and project evidence for individual technical and leadership contributions.

13. Conclusion

Trimester 1 established a strong cybersecurity foundation for PHOENIX through threat analysis, risk assessment, secure design, governance, API security work, threat detection and security testing.

Trimester 2 should build on this foundation rather than repeat it.

The main focus should now move further towards implementation, integration, testing, remediation and security validation.

My role as Cybersecurity Stream Leader will involve helping coordinate this transition while maintaining my own technical contribution through the TEAVS–ADCRS integration.

The next step is to turn the findings from this review into a clear cybersecurity work and sprint plan for Trimester 2.